Pricing and hedging of financial derivatives — Peter Tankov
2021–2022, main session
No documents/calculators allowed, exam length 2 hours

Problem: Bachelier option pricing model (12 pts)

Throughout this problem the interest rate is assumed to be zero. Let W be a one-dimensional standard Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We consider a tradable financial asset whose price can take on negative as well as positive values (e.g., a forward contract). This price will be denoted by S_t and modeled with the help of a Brownian motion with drift b and volatility $\sigma > 0$:

$$S_t = S_0 + bt + \sigma W_t.$$

We are interested in a European option on this asset with pay-off $g(S_T)$ at time T.

1. Recall the self-financing portfolio equation. Using this equation and Itô formula, show that if $u : [0,T] \times \mathbb{R} \to \mathbb{R}$ is a $C^{1,2}$ solution of the following PDE

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial S^2} = 0, \quad (t,S) \in [0,T] \times \mathbb{R} \tag{1}$$

   with terminal condition $u(T,S) = g(S)$, then there exists a self-financing portfolio, which replicates the option, and whose value V_t satisfies $V_t = u(t, S_t)$ at any time t.

2. Assuming that both V_t and $-V_t$ are admissible portfolios, argue that the price at time t of an option with pay-off g in the Bachelier model equals $u(t, S_t)$.

3. Let $\widehat{S}$ be a process defined by

$$\widehat{S}_s = S + \sigma W_{s-t}, \quad s \geq t, \tag{2}$$

   where S is a fixed constant. Let u be a $C^{1,2}$ solution of the PDE (1), such that $\frac{\partial u}{\partial S}$ is bounded. Applying the Itô formula to the function $u(s, \widehat{S}_s)$, show that

$$u(t,S) = \mathbb{E}[g(S + \sigma W_{T-t},)].$$

4. Let $g(S) = (S-K)^+$ be the pay-off of a call option. Show that the price at time t of the call option in the Bachelier model is given explicitly by

$$C_t^B(T,K,S,\sigma) = (S-K)N\left(\frac{S-K}{\sigma\sqrt{T-t}}\right) + \sigma\sqrt{T-t}n\left(\frac{S-K}{\sigma\sqrt{T-t}}\right)$$

   where n is the density function of the standard Gaussian distribution and N is its distribution function.

5. Compute the hedge ratio for call option in the Bachelier model.
6. By an arbitrage argument, show that $C_t^B(T,K,S,\sigma) \geq \underline{C} := (S-K)^+$. Also show that the classical upper bound, i.e. $C_t^B(T,K,S,\sigma) \leq S$, does not always hold in this model.
7. Compute the vega (sensitivity of the price with respect to volatility) in the Bachelier model. Show that for all $P \geq \underline{C}$, there exists a unique value σ^* such that $C_t^B(T,K,S,\sigma^*) = P$ (called Bachelier implied volatility).

Asian option in the Bachelier model We now turn our attention to the Asian option with pay-off

$$\left(\frac{1}{T}\int_0^T S_t dt - K\right)^+$$

at time T, in the model defined in the previous paragraph.

8. Let $\mathbb{Q}$ be the probability measure defined by

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = \exp\left(-\int_0^T \lambda_s dW_s - \frac{1}{2}\int_0^T \lambda_s^2 ds\right),$$

where λ is an adapted process. Show that one can choose λ such that (S_t) becomes a $\mathbb{Q}$-martingale, and given an explicit form of S in terms of a $\mathbb{Q}$-Brownian motion.

9. Recall the general formula for the price at time t of the Asian option (using the risk-neutral expectation), denoted by A_t.
10. Show that the Asian option price is given by $A_t = F\left(t, \int_0^t S_s ds, S_t\right)$ where $F(t,x,y)$ is a deterministic function.
11. Without using the explicit form of F, argue that the hedge ratio for the Asian option is given by $\delta_t^A = \frac{\partial F}{\partial y}\left(t, \int_0^t S_s ds, S_t\right)$.
12. Compute the function F and the hedge ratio in explicit form. Hint: one can use the formula

$$\mathbb{E}[(Z-m)^+] = n(m) + mN(m),$$

where $Z \sim N(0,1)$ and $m \in \mathbb{R}$.

Exercise: Quanto options (8 pts)

The goal of the exercise is to present an alternative method to price quanto options discussed during the course. Please follow the directions given rather than reproducing the arguments of the course since the problem is slightly different. In this exercise, the interest rate is not zero.

We consider an economy with two markets, domestic and foreign. The domestic interest rate is constant and denoted by r, while the foreign interest rate is constant and denoted by r^f. There are three assets: the domestic risk-free asset, with price $B_t = e^{rt}$ in domestic currency, the foreign risk-free asset with price $\widehat{B}_t^f = e^{r^f t}$ <u>in foreign currency</u> and the domestic risky asset with price denoted by S_t (in domestic currency). The exchange rate (value of one unit of foreign currency at time t expressed in domestic currency) is denoted by X_t.

We place ourselves directly under the domestic risk-neutral probability $\mathbb{Q}$ which is to be used throughout the exercise. The dynamics of the risky asset and the exchange rate under this probability are

$$\frac{dS_t}{S_t} = r dt + \sigma^S dW_t^S, \qquad \frac{dX_t}{X_t} = (r - r^f)dt + \sigma^X dW_t^X,$$

where $W_t^X = \rho W_t^S + \sqrt{1-\rho^2} W_t$ and W is a standard BM under $\mathbb{Q}$, independent from W^S, also a standard BM under $\mathbb{Q}$.

We first consider a contract with pay-off $Y_T = \frac{S_T}{X_T}$ <u>in domestic currency</u> at time T.

1. By using risk-neutral expectation, compute the price Y_t of this contract at time $t < T$. Hint: start by finding a constant γ such that

   $$\frac{S_t}{X_t} e^{\gamma t}$$

   is a $\mathbb{Q}$-martingale.

2. Give the value B_t^f and the dynamics dB_t^f of the foreign risk-free asset in domestic currency. Determine the hedging strategy for the option of the previous question using the domestic and foreign risk-free assets and the domestic risky asset. Hint: apply Itô formula to show that one can write

   $$dY_t = \delta_t dS_t + \phi_t dB_t + \psi_t dB_t^f$$

   with some processes δ_t, ϕ_t and ψ_t to be determined.

We are now interested in the <u>quanto option</u> with pay-off in domestic currency at time T given by

$$H_T = \left(\frac{S_T}{X_T} - K \right)^+ .$$

3. Show that the process (Y_t) follows the Black-Scholes model with volatility σ^Y to be determined. Argue that the option with pay-off H_T may be seen as an option on Y_T. Deduce the price of this option.

4. Determine the hedging strategy for this option using the domestic and foreign risk-free assets and the domestic risky asset.

Solution

Exercise 1

1. In the absence of interest rate, the value of a self-financing portfolio containing δ_t units of a risky asset satisfies:
$$dX_t = \delta_t dS_t.$$
Applying Itô formula, we see that $X_t = u(t, S_t)$ satisfies this dynamics with $\delta_t = \frac{\partial u}{\partial S}$.

2. Standard arbitrage argument.

3. Applying the Itô formula as suggested and making use of the PDE, we get:
$$u(T, \widehat{S}_T) = u(t, S) + \int_t^T \frac{\partial u}{\partial S}(s, \widehat{S}_s)\sigma dW_s$$
Taking the expectation of both sides and using the boundedness of $\frac{\partial u}{\partial S}$, we get the required result, after substituting the explicit form of $\widehat{S}$.

4. We compute:
$$\begin{aligned}\mathbb{E}[(S + \sigma W_{T-t} - K)^+] &= \int_{\frac{K-S}{\sigma\sqrt{T-t}}}^{\infty} (S + \sigma\sqrt{T-t}x - K)\frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}dx = \sigma\sqrt{T-t}\int_\xi^\infty (x-\xi)\frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}dx \\ &= \sigma\sqrt{T-t}\int_\xi^\infty \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} d\frac{x^2}{2} - \xi\sigma\sqrt{T-t}N(-\xi) \\ &= \sigma\sqrt{T-t}\frac{e^{-\frac{\xi^2}{2}}}{\sqrt{2\pi}} - \xi\sigma\sqrt{T-t}N(-\xi) = \sigma\sqrt{T-t}(n(\xi) - \xi N(-\xi)).\end{aligned}$$
with $\xi = \frac{K-S}{\sigma\sqrt{T-t}}$.

5. We compute the delta:
$$\frac{\partial C_t^B(T, K, S, \sigma)}{\partial S} = \mathbb{P}[S + \sigma W_{T-t} \geq K] = N\left(\frac{S-K}{\sigma\sqrt{T-t}}\right)$$

6. The first statement follows from standard arbitrage argument. The second bound does not always hold since the option price is always positive and S may be negative.

7. To compute the vega, we differentiate the option price:
$$\frac{\partial C_t^B(T, K, S, \sigma)}{\partial \sigma} = \sqrt{T-t}\, n\left(\frac{S-K}{\sigma\sqrt{T-t}}\right) > 0.$$
Furthermore, by dominated convergence, $\lim_{\sigma\to 0} C_t^B(T, K, S, \sigma) = (S-K)^+$ and by direct computation from the closed formula, $\lim_{\sigma\to\infty} C_t^B(T, K, S, \sigma) = +\infty$. This proves the existence of the unique solution.

8. Since $\widehat{W}_t := W_t + \lambda t$ is a $\mathbb{Q}$-Brownian motion, letting $\lambda = \frac{b}{\sigma}$ we get $S_t = S_0 + \sigma \widehat{W}_t$.

9. From the general theory,

$$A_t = \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{1}{T} \int_0^T S_s ds - K \right)^+ \Big| \mathcal{F}_t \right]$$

10. We compute:

$$\begin{aligned} A_t &= \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{1}{T} \int_0^t S_s ds + \frac{\sigma}{T} \int_t^T (\widehat{W}_s - \widehat{W}_t) ds + \frac{(T-t)S_t}{T} - K \right)^+ \Big| \mathcal{F}_t \right] \\ &= F\left(t, \int_0^t S_s ds, S_t\right) \end{aligned}$$

with

$$F(t,x,y) = \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{1}{T} x + \frac{(T-t)y}{T} - K + \frac{\sigma}{T} \int_t^T (\widehat{W}_s - \widehat{W}_t) ds \right)^+ \right]$$

11. We have:

$$dA_t = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial x} S_t dt + \frac{\partial F}{\partial y} dS_t + \frac{1}{2} \frac{\partial^2 F}{\partial y^2} \sigma^2 dt.$$

Since A_t is a martingale, the terms in dt cancel and we get: $dA_t = \frac{\partial F}{\partial y} dS_t$.

12. By integration by parts,

$$\int_t^T (\widehat{W}_s - \widehat{W}_t) ds = \int_t^T (T-s) dW_s,$$

so that the variance of this term equals $\frac{(T-t)^3}{3}$. Then,

$$\begin{aligned} F(t,x,y) &= \mathbb{E} \left[\left(\frac{1}{T} x + \frac{(T-t)y}{T} - K + \frac{\sigma}{T} \sqrt{\frac{(T-t)^3}{3}} Z \right)^+ \right] \\ &= \frac{\sigma}{T} \sqrt{\frac{(T-t)^3}{3}} (n(\xi) + \xi N(\xi)) \end{aligned}$$

where $Z \sim N(0,1)$ and

$$\xi = \frac{KT - x - (T-t)y}{\sigma \sqrt{\frac{(T-t)^3}{3}}}.$$

For the hedge ratio, we compute:

$$\frac{\partial F}{\partial y} = \frac{t}{T} \mathbb{P} \left[\frac{1}{T} x + \frac{(T-t)y}{T} - K + \frac{\sigma}{T} \sqrt{\frac{(T-t)^3}{3}} Z \geq 0 \right] = \frac{T-t}{T} N(-\xi).$$

Exercise 2

1. By Itô formula,

$$d\left\{\frac{S_t}{X_t}e^{\gamma t}\right\} = \frac{S_t}{X_t}e^{\gamma t}\left\{\frac{dS_t}{S_t} - \frac{dX_t}{X_t} + \gamma dt - \frac{d\langle S, X\rangle_t}{S_t X_t} + \frac{d\langle X\rangle_t}{X_t^2}\right\}$$
$$= \frac{S_t}{X_t}e^{\gamma t}\left\{rdt + \sigma^S dW_t^S - (r - r^f)dt - \sigma^X dW_t^X + \gamma dt - \sigma^X\sigma^S\rho dt + (\sigma^X)^2 dt\right\},$$

   so that by taking $\gamma = -r^f - (\sigma^X)^2 + \sigma^X\sigma^S\rho$, we obtain the martingale property. We then compute:

$$Y_t = e^{-r(T-t)}\mathbb{E}\left[\frac{S_T}{X_T}\Big|\mathcal{F}_t\right] = e^{-r(T-t)}e^{-\gamma(T-t)}\frac{S_t}{X_t}.$$

2. We have $B_t^f = X_t e^{rt}$ and therefore

$$dB_t^f = B_t^f(rdt + \sigma^X dW_t^X).$$

   Applying the Itô formula with the above choice of γ gives

$$dY_t = rY_t dt + Y_t\sigma^S dW_t^S - Y_t\sigma^X dW_t^X = \frac{Y_t}{B_t}dB_t + \frac{Y_t}{S_t}dS_t - \frac{Y_t}{B_t^f}dB_t^f.$$

3. From the previous question, the volatility of Y is given by $\sigma^Y = \sqrt{(\sigma^S)^2 + (\sigma^X)^2 - 2\rho\sigma^S\sigma^X}$. Thus the price of the quanto option is given by

$$P_t = Y_t N(d_1) - Ke^{-r(T-t)}N(d_2), \quad d_{12} = \frac{\log\frac{Y_t}{Ke^{-r(T-t)}} \pm \frac{(\sigma^Y)^2(T-t)}{2}}{\sigma^Y\sqrt{T-t}}$$

4. By Itô formula,

$$dP_t = (P_t - Y_t N(d_1))\frac{dB_t}{B_t} + N(d_1)dY_t$$